

Long-term $3+1$ simulations of primordial black hole formation during radiation domination

Submitted: August 13, 2026

Ning, Z., Cai, R., Wang, S., and Yoo, C.

Presented by: Vercil Juan
September 21, 2026

Outline

We will discuss the following:

- ▶ Preliminaries
- ▶ Primordial Black Holes
- ▶ Formulation and Numerical Setup
- ▶ Numerical Results
- ▶ Paper Conclusion



Preliminaries

Radiation Domination Epoch

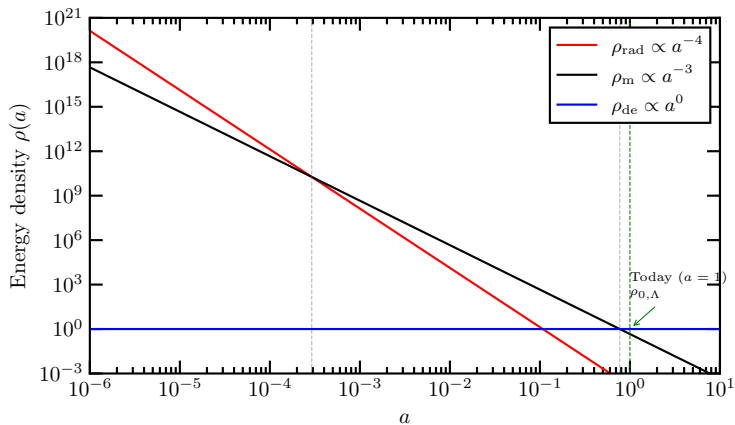


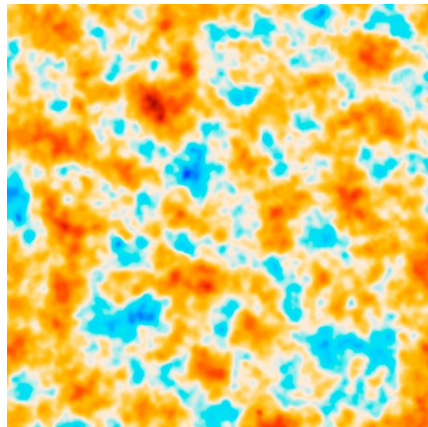
Figure: Energy density as a function of cosmic scale parameter $a(t)$

Hubble Scale

- ▶ Hubble Parameter $H = \frac{\dot{a}}{a} \sim 70 \text{ km/s/Mpc}$.
The hubble parameter, and hence all the following, are functions of time;
 $H(t)$
- ▶ Hubble Time $t_H = H^{-1} \approx 14 \text{ Gy}$
- ▶ Hubble Radius/Horizon $r_H = \frac{c}{H} \approx 4300 \text{ Mpc (14 Gly) today}$
- ▶ Hubble Mass $M_H = \frac{4\pi}{3} \rho R_H^3 = \frac{c^3}{2GH} \approx 10^{53} \text{ kg} \sim 10^{23} M_\odot$

Primordial Black Holes

- ▶ Black holes hypothesized to form in the early Universe.
- ▶ Formed from the direct gravitational collapse of dense matter and radiation in the first fractions of a second after the big-bang, rather than the late-time collapse of stars.
- ▶ Can form from sufficiently large primordial curvature perturbations.
- ▶ The paper studies PBH formation during radiation domination.



Primordial Black Holes

Can possibly explain/add information to

- ▶ Dark matter
- ▶ Early gravitational waves
- ▶ Seeds of supermassive blackholes
- ▶ Probe early universe physics

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What the paper aims to do

- ▶ Perform a fully $3 + 1$ dimension numerical relativity simulation of PBH formation
- ▶ Starting with formation, accretion, and following through its evolution over a long time (order of several Hubble times)
- ▶ Determine the critical formation parameters, including the threshold perturbation amplitude μ_c and the critical exponent γ governing the PBH mass scaling.

Primordial Black Holes

Basic Picture of PBH Formation

- ▶ After inflation, the Universe is radiation-dominated, and superhorizon curvature perturbations $\zeta(\mathbf{x})$ are frozen into the initial conditions.
- ▶ While a mode is superhorizon ($\lambda > c/H$), causality prevents it from evolving — its amplitude stays fixed.
- ▶ As the Universe expands, the comoving Hubble radius $c/(aH)$ grows. Once $\lambda = c/H$, the mode *re-enters the horizon* and starts to evolve dynamically.

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- ▶ As the Universe expands, the comoving Hubble radius $c/(aH)$ grows. Once $\lambda = c/H$, the mode *re-enters the horizon* and starts to evolve dynamically.
- ▶ The overdense region then faces a competition:
 - ▶ pressure gradients push the fluid outward,
 - ▶ self-gravity pulls it inward.
- ▶ If the amplitude μ is large enough, $\mu > \mu_c$, gravity wins and the region collapses, an *apparent horizon* forms, signalling the birth of a PBH.
- ▶ Perturbations below the threshold instead disperse, leaving no black hole behind.

Formulation

Spacetime Evolution

Write the spacetime metric in the standard $3 + 1$ form

$$ds^2 = -a^2 dt^2 + \gamma_{ij}(dx^i + \beta^i)(dx^j + \beta^j t) \quad (1)$$

where a , β^i , and γ_{ij} denote the lapse function, shift vector, and spatial metric, respectively.

The future-directed unit vector normal to a constant- t hypersurface is

$$n_\mu = (-\alpha, 0, 0, 0)$$

Formulation

Spacetime Evolution

Define the extrinsic curvature as

$$K_{ij} = -\frac{1}{2\alpha}(\partial_t \gamma_{ij} - \mathcal{L}_\beta \gamma_{ij}) \quad (2)$$

Evolve the Einstein equations using the Baumgarte-Shapiro-Shibata-Nakamura (BSSN) formulation.

- ▶ a widely used reformulation of Einstein's equations in numerical relativity that enables stable, long-term simulations of black hole and neutron star spacetimes.

Formulation

Spacetime Evolution

The evolved geometric variables are:

$$\chi \equiv \gamma^{-1/3}, \quad \tilde{\gamma}_{ij} \equiv \chi \gamma_{ij}, \quad K \equiv \gamma^{ij} K_{ij},$$
$$\tilde{A}_{ij} \equiv \chi \left(K_{ij} - \frac{1}{3} \gamma_{ij} K \right), \quad \tilde{\Gamma}^i \equiv \tilde{\gamma}^{jk} \tilde{\Gamma}_{jk}^i.$$

The matter source terms entering the equations are given by the Eulerian projections

$$E \equiv n_\mu n_\nu T^{\mu\nu} \quad S_i \equiv -\gamma_{i\mu} n_\nu T^{\mu\nu}$$
$$S_{ij} = \gamma_{i\mu} \gamma_{j\nu} T^{\mu\nu} \quad S = \gamma^{ij} S_{ij}$$

The Hamiltonian and momentum constraints take the form

$$\mathcal{H} \equiv {}^{(3)}R + K^2 - K_{ij} K^{ij} - 16\pi E = 0$$
$$\mathcal{M}_i \equiv D_j K^j_i - D_i K - 8\pi S_i = 0$$

Formulation

For the radiation dominated Universe, they describe the matter sector as a perfect fluid, with the stress energy tensor as

$$T^{\mu\nu} = (\rho + p)u^\mu u^\nu + pg^{\mu\nu} \quad (3)$$

- ▶ ρ is the proper energy density measured in the fluid rest frame
- ▶ p is the pressure
- ▶ u^μ is the fluid four-velocity

They adopt a linear barotropic equation of state

$$p = \omega \rho$$

and set $\omega = \frac{1}{3}$ throughout the simulation.

Primordial Curvature Perturbation

The paper uses a spherical Gaussian curvature perturbation

$$\zeta(r) = \mu e^{-k^2 r^2} W(r). \quad (4)$$

- ▶ μ : amplitude of the perturbation.
- ▶ k^{-1} : characteristic length scale.
- ▶ $W(r)$: smooth cutoff near the numerical boundary.

Primordial Curvature Perturbation

Window function

$$W(r) = \begin{cases} 1, & 0 \leq r \leq r_W \\ 1 - \frac{[(L-r_W)^6 - (L-r)^6]^6}{(L-r_W)^{36}}, & r_W < r < L \\ 0, & r \geq L \end{cases}$$

- ▶ L is the half side length of the cubic simulation
- ▶ r_W is the radius at which the window begins to taper off.

Critical Mass Scaling

Near the threshold, the PBH mass obeys

$$M_{\text{BH}} = M_H K (\mu - \mu_c)^\gamma.$$

Here

- ▶ M_H : Hubble mass,
- ▶ K : proportionality constant,
- ▶ μ_c : critical perturbation amplitude,
- ▶ γ : critical exponent.

Numerical Structure

- ▶ The paper implements the numerical scheme in the open-source numerical-relativity code GRChombo, which employs the Chombo library for block adaptive mesh refinement (AMR)
- ▶ Incorporate the flux-conservative fluid scheme described above into the existing BSSN evolution infrastructure
- ▶ Differentiate the BSSN variables using fourth order finite-difference stencils and update the fluid conserved variables through numerical fluxes at cell faces
- ▶ Integrate the coupled system using the method of lines and the classical fourth-order Runge–Kutta scheme.

Numerical Structure

- ▶ **Domain:** evolve on the octant $0 \leq x, y, z \leq L$ with reflection boundaries — equivalent to a periodic box of side $2L$ for the reflection-symmetric configurations considered here, at $\sim 8\times$ lower cost; validated against a full periodic run at the same resolution.
- ▶ **Resolution:** base grid 64^3 cells on the octant, five AMR levels with refinement ratio 2 giving $\Delta x_{\min} = L/2048$, refined on gradients of χ and E ; for the long accretion runs, $L \rightarrow 2$ and the base grid is doubled so the physical resolution and AMR criteria are unchanged.

Simulation

Setup

- ▶ The simulation begins with a curvature perturbation $\zeta(\mathbf{x})$ placed *well outside* the Hubble horizon.
- ▶ Enforced by the small parameter

$$\epsilon \equiv \frac{k}{a_i H_i} \ll 1,$$

i.e. physical wavelength $\gg$ Hubble radius.

- ▶ In this limit, spatial gradients are suppressed by powers of ϵ — causality freezes the perturbation's evolution.
- ▶ The initial slice is therefore clean, unambiguous, and physically motivated: this is what inflation produces.

Simulation

- ▶ Because $\epsilon \ll 1$, the geometric variables can be expanded order-by-order in ϵ (Shibata–Sasaki).
- ▶ Through $\mathcal{O}(\epsilon^2)$:

$$\chi = a_i^{-2} e^{-2\zeta} \left[1 + \frac{2f}{3(1+\omega)} \frac{1}{(a_i H_i)^2} \right] + \mathcal{O}(\epsilon^4),$$

and similarly for $\tilde{\gamma}_{ij}$, K , $\tilde{A}_{ij}$, α .

- ▶ The fluid variables are then recovered by *solving the Hamiltonian and momentum constraints* — not from the truncated expressions.
- ▶ Result: constraint-satisfying initial data that exactly represents a superhorizon perturbation.

Simulation

- ▶ The perturbation does *not* move — its comoving wavelength $\lambda_{\text{com}} = 2\pi/k$ is constant.
- ▶ What grows is the *comoving Hubble radius* $c/(aH)$, because a grows and H decreases.
- ▶ Horizon entry occurs when the two meet:

$$a(t_H) r_m = H^{-1}(t_H),$$

defining $t_H = t_i (a_i H_i r_m)^{1/(1-\alpha_\omega)}$ and $M_H = t_H/(2\alpha_\omega)$.

- ▶ After t_H , the perturbation is causally connected and begins to evolve dynamically.

Simulation

- ▶ Once subhorizon, the overdense region faces two opposing forces:
 - ▶ **Pressure gradients** push the radiation fluid outward.
 - ▶ **Self-gravity** pulls the region inward.
- ▶ The outcome depends on the perturbation amplitude μ :

$$\begin{cases} \mu < \mu_c : & \text{dispersal, no PBH,} \\ \mu > \mu_c : & \text{collapse, PBH forms.} \end{cases}$$

- ▶ The threshold μ_c depends on the equation of state and the shape of the curvature profile.
- ▶ Just above threshold, PBH mass obeys a critical scaling law
 $M_{\text{BH}} \propto (\mu - \mu_c)^\gamma$.

Simulation

- ▶ When collapse succeeds, an *apparent horizon* forms — the operational signature of PBH birth.
- ▶ The mass is measured from the horizon area:

$$M_{\text{BH}} = \sqrt{A_{\text{AH}}/16\pi}.$$

- ▶ The paper then follows the PBH for *many Hubble times* after formation, tracking its accretion of the surrounding radiation.
- ▶ The late-time growth is fit to the Zel'dovich–Novikov law:

$$\dot{M}_{\text{BH}} = \frac{3F}{2} \frac{M_{\text{BH}}^2}{t^2}.$$

- ▶ This is the “long-term” part of the title: the simulation spans horizon entry, collapse, and post-formation accretion.

Cosmological Gauge

- ▶ Standard $1 + \log$ slicing drives α away from unity on an expanding background ($K = -3H < 0$). Fix: background-subtracted slicing $(\partial_t - \beta^i \partial_i) \alpha = -2\alpha(K - K_b)$, with $K_b = -3H(t)$.
- ▶ Physical coordinate speeds redshift as a^{-1} , so the CFL limit would permit $\Delta t \propto a(t) \Delta x$.
- ▶ But the Gamma-driver's gauge modes do *not* redshift: they see the *conformal* metric $\tilde{\gamma}_{ij} = \delta_{ij}$, so their characteristic speeds $v_L = \sqrt{4b/3}$ stay $O(1)$ in coordinate units.
- ▶ Result: a fixed time step, and an increasingly conservative CFL number as the Universe expands.
- ▶ Scale the Gamma-driver coefficients with $a(t)$:

$$b(t) = b_0 \left[\frac{a_i}{a(t)} \right]^2, \quad \eta_{\text{GD}}(t) = \eta_{\text{GD},0} \frac{a_i}{a(t)},$$

with $b_0 = 3/4$, $\eta_{\text{GD},0} = 1$.

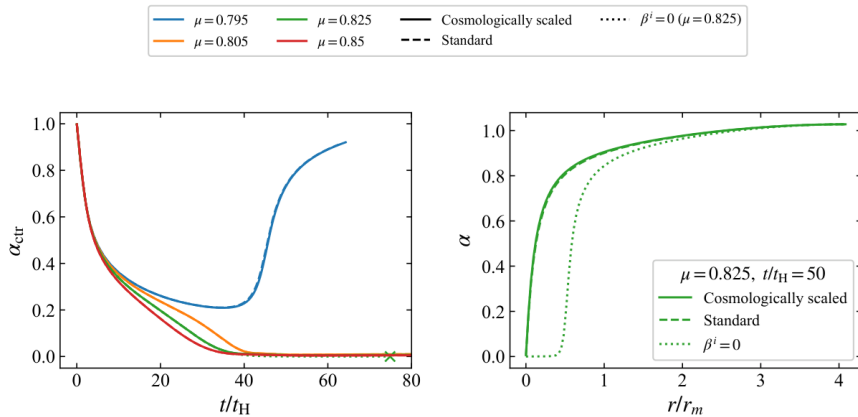
Cosmological Gauge

- ▶ Then the two stability parameters stay constant when $\Delta t \propto a \Delta x$:

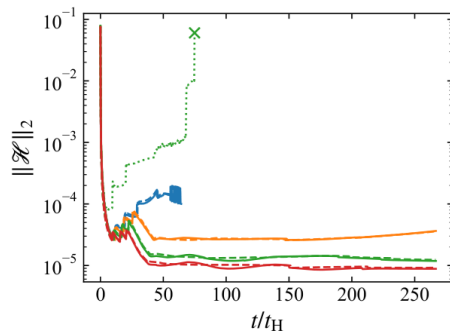
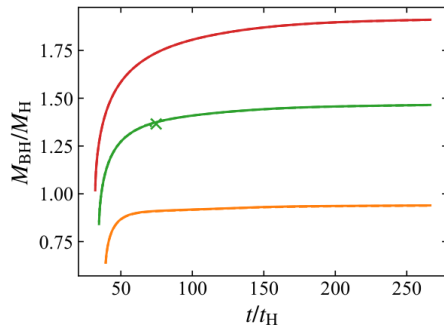
$$C_\beta \propto \frac{\Delta t}{a \Delta x} = \text{const}, \quad C_\eta \propto \frac{\Delta t}{a} = \text{const}.$$

- ▶ The driver does *not* turn off: proper-distance gauge speed $a v_L$ stays constant; only the convergence rate in cosmic time changes.
- ▶ **Independent check:** a conformal-time gauge ($\alpha_\eta = a\alpha_t$, $\beta_\eta^i = a\beta_t^i$) where physical speeds are already constant, so the standard driver with $b = 3/4$, $\eta_{\text{GD}} = 1$ works directly.
- ▶ **Payoff:** Δt grows to $\Delta t/t_H \simeq 0.52$ (vs. standard cap 2.60×10^{-3}) — a $\sim 200\times$ larger step and a $\sim 94\times$ reduction in coarse advances, with identical AH masses and constraints.

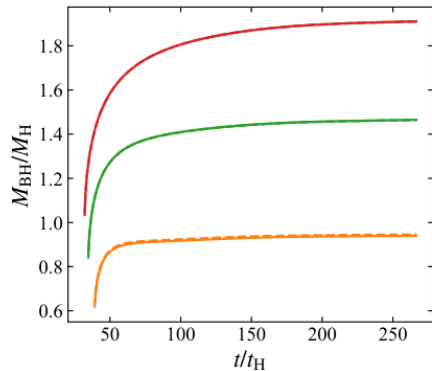
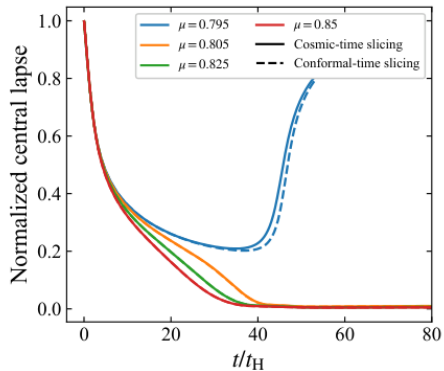
Results



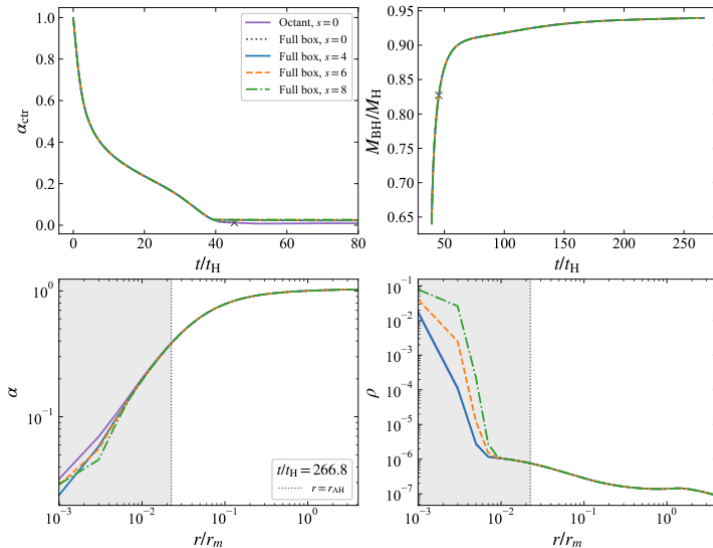
Results



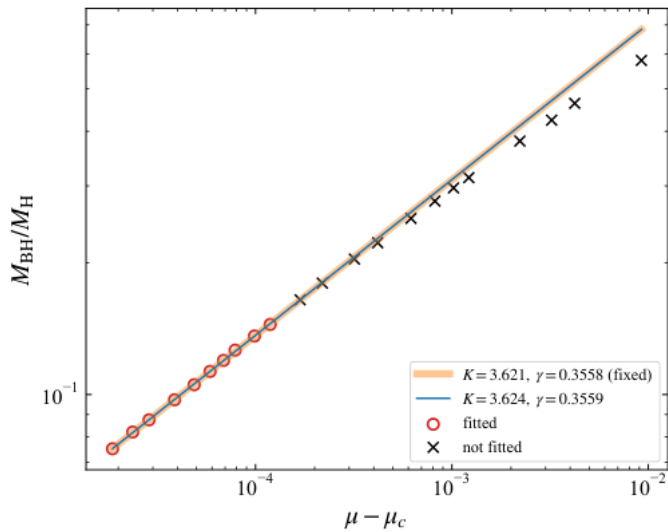
Results



Results



Results



Critical Exponent

The simulation finds

$$\gamma \simeq 0.3559.$$

Therefore,

$$M_{\text{BH}} \propto (\mu - \mu_c)^{0.3559}.$$

As

$$\mu \rightarrow \mu_c^+,$$

the PBH mass approaches zero in the critical-scaling regime.

Critical Amplitude

The simulation finds the collapse threshold

$$0.79578 < \mu_c < 0.79580$$

refining earlier 3D results and matching the spherically symmetric Misner–Sharp value $\mu_c \simeq 0.79579 \pm 1 \times 10^{-5}$ — a strong validation of the 3D fluid evolution.

Conclusion

- ▶ Developed a 3D numerical-relativity framework for PBH formation from superhorizon curvature perturbations during radiation domination.
- ▶ Cosmologically scaled Gamma-driver gives $\Delta t \propto a \Delta x$ — $\sim 94\times$ fewer coarse advances, identical physics.
- ▶ Threshold $0.79578 < \mu_c < 0.79580$, matching Misner–Sharp; critical exponent $\gamma \simeq 0.3559$, matching the radiation value.
- ▶ Late-time accretion over many Hubble times fits the Zel’dovich–Novikov law.
- ▶ First 3D determination of both the critical scaling relation and the post-formation accretion law.

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- [2] Carr, B. J. (1975). The primordial black hole mass spectrum. *The Astrophysical Journal*, 201, 1–19. <https://doi.org/10.1086/153853>
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